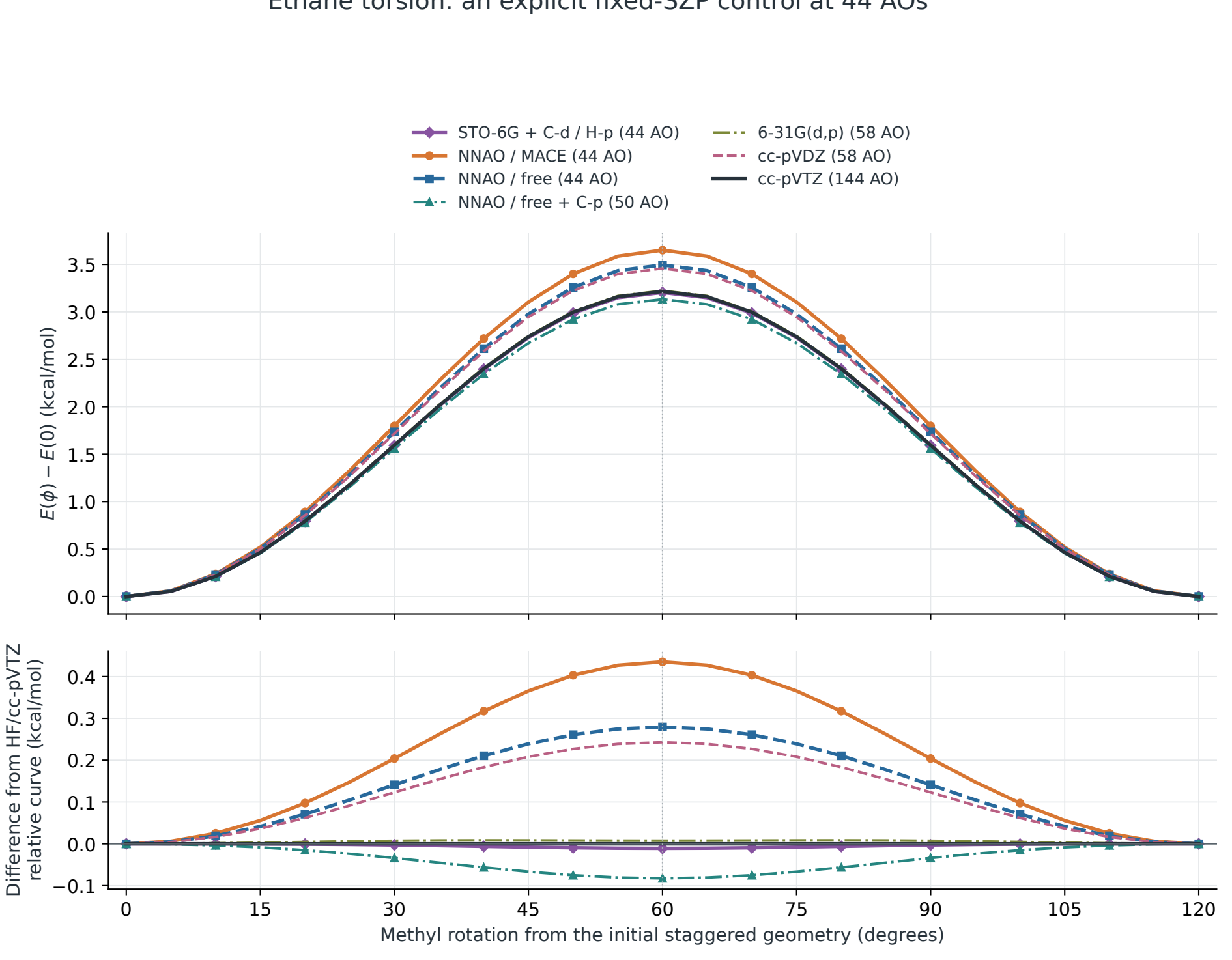
Ethane torsion: an explicit fixed-SZP control at 44 AOs
STO-6G + C-d / H-p (44 AO)
NNAO / MACE (44 AO)
NNAO / free (44 AO)
NNAO / free + C-p (50 AO)
6-31G(d,p) (58 AO)
cc-pVDZ (58 AO)
cc-pVTZ (144 AO)
$E(\phi) - E(0)$ (kcal/mol)
Difference from HF/cc-pVTZ relative curve (kcal/mol)
Methyl rotation from the initial staggered geometry (degrees)
25 rigid geometries; DF-RHF / def2-universal-jkfit, spherical AOs. NNAO MACE update 3351.
Fixed control: unchanged STO-6G + C-d exponent 0.8 + H-p exponent 1.1 (Bohr^-2).
Each relative curve uses its own zero. HF/cc-pVTZ is a comparison, not an exact reference.